

# How Wasteful is Signaling?\*

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## Abstract

Signaling is wasteful. But how wasteful? We study the fraction of surplus dissipated in a separating equilibrium. For isoelastic environments, this waste ratio has a simple formula:  $\beta/(\beta + \sigma)$ , where  $\beta$  is the benefit elasticity (reward to higher perception) and  $\sigma$  is the strain elasticity (higher types' cost advantage). The ratio is constant across types and independent of other parameters, including cost convexity. A constant waste ratio characterizes the isoelastic class. An application to winner-take-all signaling tournaments shows that with  $N$  candidates, exactly  $(N - 1)/N$  of the surplus dissipates—the same as in Tullock contests.

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# 1. Introduction

Signaling is wasteful. In the canonical [Spence \(1973\)](#) model and its innumerable applications and descendants, agents take costly actions to distinguish themselves from lower types. The resulting separating equilibrium reveals information but necessarily dissipates surplus—a fundamental source of inefficiency under asymmetric information.<sup>1</sup>

But *how* wasteful is signaling, and what does it depend on? Despite more than 50 years of research, the literature offers no simple answer.

A natural intuition suggests that the magnitude of waste should depend on the *difficulty* of signaling. If signaling costs are highly convex in the action (a “hard” test), agents encounter high marginal costs quickly, which ought to limit total expenditure. This reasoning suggests that policies that raise stakes would exacerbate waste, while policies that make signaling more difficult—via exam difficulty, advertising costs, or certification requirements—could reduce it. Similar, one might expect that limiting the signaling *stakes*—the (marginal) benefit from inducing a higher market perception—would also help.

Our paper shows that this intuition is incomplete, even misleading. We study the classic continuum-type signaling model used in economics, presented formally in [Section 2](#), and focus on the essentially unique (fully-)separating equilibrium. Our main result, [Theorem 1](#) in [Section 3](#), has two parts. First, under a standard multiplicative cost structure, the proportion of surplus burned by any type—its *waste ratio*—is invariant to both difficulty and stakes. Importantly, difficulty captures not just the scale of costs, but also the shape (convexity). Difficulty and stakes both affect which actions agents take in equilibrium, and stakes can also affect the equilibrium costs, but neither affects the waste ratio.

Second, within a canonical isoelastic class of costs and benefits, we obtain a *constant of dissipation*: the waste ratio is the constant

$$W = \frac{\beta}{\beta + \sigma}.$$

Here,  $\beta > 0$  is the elasticity of benefits (how steeply rewards rise with perceived type) and  $\sigma > 0$  is the elasticity of cost strain (how quickly higher types’ comparative advantage grows). Waste depends only on  $\beta/\sigma$ , increasing from zero to one in that ratio. To reduce waste, policies must either flatten the reward structure or steepen the cost strain—not tackle

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<sup>1</sup> Of course, signaling activities can also generate benefits: education builds human capital; and prosocial behavior brings positive externalities, which signaling can amplify ([Bénabou and Tirole, 2006](#)). Our paper focuses on the wasteful component of signaling.

difficulty or stakes. The constant waste ratio provides a simple and interpretable answer to our motivating question; among other things, it avoids issues of aggregation across types.

We then show in [Theorem 2](#) that (under multiplicative costs) waste is constant across types if and only if the costs and benefits satisfies a constant relative elasticity condition. Hence, up to the labeling of types, the isoelastic class is the unique setting for such uniform dissipation. This characterization provides a theoretical foundation for the isoelastic specification.

In [Section 4](#), we apply the waste formula to signaling in tournaments. Under suitable assumptions, a winner-take-all market with  $N$  candidates competing for a prize has benefit elasticity  $\beta = N - 1$  and strain elasticity  $\sigma = 1$ , yielding a waste ratio of  $(N - 1)/N$ . Thus, the fraction of surplus burned increases in  $N$ , converging to full dissipation as  $N \rightarrow \infty$ . The  $(N - 1)/N$  formula is precisely the rent dissipation rate in symmetric Tullock lottery contests, revealing an unexpected connection between signaling and contest theory.

The conclusion, [Section 5](#), discusses implications, interpretations, and limitations.

**Related Literature.** The costly signaling literature in economics, surveyed by [Riley \(2001\)](#) and [Sobel \(2009\)](#), emphasizes the conditions for separating equilibria and that information revelation entails surplus dissipation. However, we are aware of little work that systematically quantifies this waste. One exception is [Bernheim and Bodoh-Creed \(2023\)](#), who show that dissipation can vanish when agents have heterogeneous bliss points and choose a large number of actions. Our paper quantifies waste in the canonical signaling setting (with homogeneous bliss points), when distortions are non-negligible.

In the biological signaling literature, [Nöldeke and Samuelson \(1999\)](#) show that offspring's equilibrium cost is proportional to parents' fitness loss, with a constant depending only on genetic relatedness. Their analysis does not yield a constant waste ratio (cost relative to sender's benefit, which need not track parental loss), and their assumption of a linear cost precludes questions about signaling difficulty. But our paper shares with them a common theme that dissipation can admit a simple formula based on primitive parameters.

We discuss some other literature connections later in the paper.

## 2. Model

An agent has type  $\theta \in \Theta := [0, \bar{\theta}]$ , where  $0 < \bar{\theta} \leq \infty$ .<sup>2</sup> The type is drawn from a continuous cumulative distribution  $F$  with support  $\Theta$ . After privately learning his type, the agent chooses a publicly observable signal or action  $a \in \mathbb{R}_{\geq 0}$ . An observer sees the action and forms her expectation  $\hat{\theta}$  of the agent's type. The agent's payoff is  $V(\hat{\theta}) - C(a, \theta)$ . We maintain throughout the following assumption (primes and subscripts on functions denote derivatives in the usual manner).

**Assumption 1.** *The benefit function  $V : \Theta \rightarrow \mathbb{R}_{\geq 0}$  and cost function  $C : \mathbb{R}_{\geq 0} \times \Theta \rightarrow \mathbb{R}_{\geq 0} \cup \{\infty\}$  respectively satisfy:*

1.  *$V$  is twice continuously differentiable, with  $V(0) = 0$  and  $V'(\theta) > 0$  for all  $\theta > 0$ .*
2. *On  $\mathbb{R}_{\geq 0} \times (0, \bar{\theta})$ ,  $C$  is finite and continuous with  $C(0, \theta) = 0$ ; on  $\mathbb{R}_{> 0} \times (0, \bar{\theta})$ ,  $C$  is continuously differentiable with  $C_a > 0$ , and  $C_a$  is continuously differentiable with  $C_{a\theta} < 0$ . The lowest type has cost  $C(a, 0) = \lim_{\theta \downarrow 0} C(a, \theta)$  for all  $a$ .*

Part 1 of [Assumption 1](#) says that agents prefer to be perceived as higher types, with the benefit from the lowest perception normalized to zero. Part 2 says that higher actions are costlier, and higher types have lower marginal costs. While it would be natural for costs to be convex in the action, we don't need to assume that. Part 2 also normalizes  $C(0, \theta) = 0$  for all  $\theta$ , so that the payoff from taking the lowest action and receiving the lowest perception is zero. The technical conditions in the two parts are largely standard; note that we allow for type 0 to have infinite costs for actions  $a > 0$  to encompass canonical isoelastic costs, detailed in [Section 3](#).

**Equilibrium.** We study (fully-)separating equilibria. The equilibrium definition is standard and relegated to [Appendix A](#), where [Proposition 1](#) shows that any separating equilibrium can be described by a continuous, strictly increasing agent (pure) strategy  $A : \Theta \rightarrow \mathbb{R}_{\geq 0}$  that is differentiable for  $\theta > 0$  and satisfies  $A(0) = 0$ . Incentive compatibility requires that each type  $\theta$  optimally chooses  $A(\theta)$  given that the observer correctly inverts the strategy on the equilibrium path, i.e., when beliefs satisfy  $\hat{\theta}(a) = A^{-1}(a)$  for  $a \in [0, A(\bar{\theta})]$ . Off-path beliefs can simply be set to  $\hat{\theta}(\cdot) = 0$ .

Thus, in a separating equilibrium  $A$ , any type  $\theta$  solves

$$\max_a [V(\hat{\theta}(a)) - C(a, \theta)],$$

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<sup>2</sup> We use the notation  $[0, x]$  to mean  $[0, x]$  if  $x < \infty$  and  $[0, \infty)$  if  $x = \infty$ .

where  $\hat{\theta}(\cdot) = A^{-1}(\cdot)$ . For  $\theta > 0$ , the first-order condition evaluated at the optimal action  $A(\theta)$  is

$$C_a(A(\theta), \theta) = V'(\theta) \cdot \hat{\theta}'(A(\theta)) = \frac{V'(\theta)}{A'(\theta)}, \quad (1)$$

where the first equality uses  $\hat{\theta}(A(\theta)) = \theta$  and the second uses  $\hat{\theta}'(A(\theta)) = 1/A'(\theta)$ .

[Equation 1](#) has a simple interpretation: its left-hand side is the marginal cost of increasing the action; the right-hand side is the marginal benefit of inducing a higher belief scaled by the marginal increase in action required for that higher belief. Together with  $A(0) = 0$ , [Equation 1](#) defines a boundary-value differential equation. There is a unique solution by standard arguments.<sup>3</sup> That solution, which we continue to refer to as just  $A$  subsequently, characterizes the unique separating equilibrium (uniqueness is up to the specification of off-path beliefs); sufficiency is verified by [Proposition 2](#) in [Appendix A](#).

**The Waste Ratio.** To measure signaling inefficiency—how much surplus is dissipated by costly signaling—we define three quantities. The opt-out payoff  $U^O(\theta) := V(0) - C(0, \theta) = 0$  is what a type would get if it chose the least-cost action and was perceived as the lowest type. The complete-information payoff  $U^{CI}(\theta) := V(\theta) - C(0, \theta) = V(\theta)$  is what a type would get if it revealed itself costlessly. Lastly,  $U(\theta) := V(\theta) - C(A(\theta), \theta)$  is a type’s separating equilibrium payoff.

**Definition 1.** The *waste ratio* for type  $\theta$  is the fraction of its payoff from costless separation that is dissipated through costly signaling:

$$W(\theta) := \frac{U^{CI}(\theta) - U(\theta)}{U^{CI}(\theta) - U^O(\theta)} = \frac{C(A(\theta), \theta)}{V(\theta)}.$$

We refer to the denominator  $V(\theta)$  as *surplus*: it is the maximum payoff type  $\theta$  could secure if types were verifiable at zero cost.<sup>4</sup> The surplus interpretation is motivated by thinking of  $V(\theta)$  as the social value of the agent, with a competitive market paying the agent that value. The numerator  $C(A(\theta), \theta)$  is the deadweight loss incurred to credibly secure that surplus. The ratio  $W(\theta)$  thus measures the effective “tax” the separating equilibrium imposes on the value of the underlying information.

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<sup>3</sup> More precisely, standard existence and uniqueness results for ordinary differential equations can be applied on  $(0, \bar{\theta})$  and extended to the lower boundary by continuity; see the arguments in, for example, [Mailath \(1987\)](#) or [Kartik \(2009\)](#).

<sup>4</sup> Since  $V(0) = 0$ , the waste ratio for the lowest type,  $W(0)$ , is ill-defined. We will ignore that; our formal results address  $W(\theta)$  for all  $\theta > 0$ .

Our goal is to study this waste ratio, which, to our surprise, appears to be a novel object of general analysis in signaling games.<sup>5</sup>

### 3. The Constant of Dissipation

For some discussion below, it is useful to view  $V(\theta) = s \cdot B(\theta)$ , where  $s > 0$  represents the agent's *stakes* in signaling. (We interchangeably refer to both  $V$  and  $B$  as the benefit function.) We also focus on multiplicatively separable costs that are commonplace in signaling models. Formally, we assume hereafter that

$$C(a, \theta) = D(a) \cdot S(\theta), \quad (2)$$

where  $D : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$  and  $S : \Theta \rightarrow \mathbb{R}_{>0} \cup \{\infty\}$ . Here  $D(a)$  represents the *difficulty* of action  $a$  and  $S(\theta)$  represents the *strain* experienced by type  $\theta$ . Given  $S(\cdot) > 0$  (merely a sign convention), [Assumption 1](#) part 2 implies (i)  $D(0) = 0$  and  $D'(a) > 0$  for  $a > 0$ ; and (ii) for  $\theta > 0$ , we have  $S(\theta)$  finite and  $S'(\theta) < 0$ , while  $S(0) = \lim_{\theta \downarrow 0} S(\theta)$ . Note that  $S(0) = \infty$  corresponds to type 0 facing prohibitive signaling costs for any  $a > 0$ .<sup>6</sup>

The following class of costs and benefits are central to our results.

**Definition 2.** An *isoelastic environment* is defined by

$$B(\theta) = \theta^\beta \quad \text{and} \quad S(\theta) = \theta^{-\sigma},$$

where  $\beta > 0$  is the *benefit elasticity*, and  $\sigma > 0$  is the *strain elasticity*.

Note that the definition stipulates isoelasticity in  $B$  and  $S$ , but not  $D$ . Our first result says that multiplicative costs ensure the waste ratio is independent of stakes and difficulty, and isoelasticity further implies a constant waste ratio across types.

**Theorem 1.** Assume the multiplicative costs (2). Then:

1. The waste ratio  $W(\theta)$  is invariant to stakes ( $s$ ) and difficulty ( $D(\cdot)$ ).

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<sup>5</sup> The waste ratio can be viewed as analogous to the “Price of Anarchy” in algorithmic game theory ([Koutsoupias and Papadimitriou, 1999](#); [Roughgarden, 2005](#)). That literature generally studies worst-case bounds across multiple equilibria; we are interested in the exact value in the separating equilibrium. Furthermore, we define waste pointwise across types, whereas Bayesian Price of Anarchy typically uses ex-ante expected payoffs ([Roughgarden, Syrgkanis and Tardos, 2017](#)). A consequence of our results is that the latter distinction is rendered moot in [isoelastic environments](#).

<sup>6</sup> But  $C(0, 0) = D(0)S(0) = 0$ , using the convention  $0 \times \infty = 0$ .

2. In an isoelastic environment, the waste ratio is constant: for any  $\theta > 0$ , it is

$$W(\theta) = \frac{\beta}{\beta + \sigma}. \quad (3)$$

Equation 3 is a remarkably simple formula for how much surplus is wasted by signaling. The textbook example (e.g. Fudenberg and Tirole, 1991, p. 329) with  $B(\theta) = \theta$  and  $C(a, \theta) = a/\theta$  corresponds to  $\beta = \sigma = 1$ , and so it is precisely 50% of the surplus that is dissipated there. More generally, only the ratio  $\beta/\sigma$  matters; waste is monotonically increasing in  $\beta/\sigma$ , ranging all the way from 0 to 1.

These directional effects are intuitive. Higher  $\beta$  means a greater incentive to separate from lower types; the rat race for higher beliefs becomes fiercer and more of the surplus is burned. Conversely, higher  $\sigma$  confers a stronger relative cost advantage to higher types, so separation requires less waste. What is surprising in Theorem 1 is the irrelevance of stakes  $s$  and difficulty  $D(\cdot)$ . To explain that irrelevance and how isoelasticity delivers a constant waste, we present the theorem's straightforward proof.

**Proof of Theorem 1.** Substituting  $C_a(a, \theta) = D'(a)S(\theta)$  and  $V(\theta) = sB(\theta)$  into Equation 1, the separating strategy  $A$  satisfies (for  $\theta > 0$ ) the differential equation

$$D'(A(\theta))A'(\theta) = \frac{sB'(\theta)}{S(\theta)}.$$

As the left-hand side is  $\frac{d}{d\theta}D(A(\theta))$ , integrate from 0 to  $\theta$  to obtain

$$D(A(\theta)) = s \int_0^\theta \frac{B'(t)}{S(t)} dt,$$

using the fundamental theorem of calculus and  $D(A(0)) = D(0) = 0$ . Since  $C(A(\theta), \theta) = D(A(\theta))S(\theta)$ , the waste ratio is

$$W(\theta) = \frac{D(A(\theta))S(\theta)}{V(\theta)} = \frac{S(\theta)}{B(\theta)} \int_0^\theta \frac{B'(t)}{S(t)} dt. \quad (4)$$

Both the stakes  $s$  and the difficulty  $D(\cdot)$  have canceled, establishing part 1. For part 2, substituting the isoelastic form  $B(\theta) = \theta^\beta$  and  $S(\theta) = \theta^{-\sigma}$  into (4) and simplifying yields

$$W(\theta) = \theta^{-\sigma-\beta} \int_0^\theta \beta t^{\beta-1+\sigma} dt = \theta^{-\beta-\sigma} \frac{\beta}{\beta + \sigma} \theta^{\beta+\sigma} = \frac{\beta}{\beta + \sigma}. \quad \square$$

While [Theorem 1](#) shows invariance to both stakes and difficulty, there is a sense in which the latter runs deeper. Changes in difficulty do not affect equilibrium costs: agents adjust their actions to any change in  $D$ —whether scale or shape—exactly enough to keep equilibrium costs unchanged.<sup>7</sup> Stakes operate differently: higher stakes do increase equilibrium costs, but benefits rise proportionally, leaving the waste ratio unchanged. The following example illustrates.

**Example 1.** An isoelastic environment with difficulty  $D(a) = a^\gamma$  for  $\gamma > 0$  yields the following separating equilibrium quantities:

$$A(\theta) = \left( \frac{s\beta}{\beta + \sigma} \right)^{1/\gamma} \theta^{(\beta+\sigma)/\gamma} \quad \text{and} \quad C(A(\theta), \theta) = \frac{s\beta}{\beta + \sigma} \theta^\beta.$$

The difficulty convexity parameter  $\gamma$  affects equilibrium actions, but not equilibrium cost, and hence not waste. Stakes  $s$  affect actions and cost, but not waste.  $\diamond$

Multiplicative separability of costs is important for [Theorem 1](#) part 1; [Example 2](#) in [Appendix C](#) confirms that more generally the waste ratio can depend on stakes.<sup>8</sup> Furthermore, the isoelastic environment is also important for part 2 of the theorem. In fact, up to a normalization of types, the constant-waste property characterizes isoelasticity under multiplicative costs. That is the content of our next result, whose proof is in [Appendix B](#).

**Theorem 2.** *Assume the multiplicative costs (2). The waste ratio  $W(\theta)$  is constant for all  $\theta > 0$  if and only if the benefit function  $V$  and the strain function  $S$  satisfy*

$$-\frac{d \ln S(\theta)}{d \ln V(\theta)} = \rho \tag{5}$$

*for some constant  $\rho > 0$ . In that case, the waste ratio is  $W(\theta) = 1/(1 + \rho)$ .*

[Equation 5](#) is a constant relative elasticity condition: the (log) rate at which strain declines must be proportional to the (log) rate at which benefits increase. When this proportionality holds uniformly across types, the tension between signaling incentives and costs resolves identically for everyone.

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<sup>7</sup> This is seen in [Equation 4](#), noting that  $C(A(\theta), \theta) = sB(\theta)W(\theta)$ .

<sup>8</sup> Multiplicative costs are immaterial for another invariance: the waste ratio  $W(\theta)$  does not depend on the type distribution  $F$ . This invariance owes to the well-known property that the separating equilibrium strategy only depends on the support of  $F$ . The strategy discontinuity at complete information carries over to waste; in particular, under isoelasticity, waste equals  $\beta/(\beta + \sigma)$  for any full-support  $F$ , even though it would be zero under complete information.



To see why [Theorem 2](#) characterizes the isoelastic environment up to relabeling types, note that integrating [Equation 5](#) yields  $S = \kappa V^{-\rho}$  for some  $\kappa > 0$ . Since  $V$  is strictly increasing and  $V(0) = 0$ , it can be reparametrized as  $V(\tilde{\theta}) = s\tilde{\theta}^\beta$  via the change of variables  $\tilde{\theta} = (V(\theta)/s)^{1/\beta}$ , which then gives  $S(\tilde{\theta}) = s^{-\rho}\tilde{\theta}^{-\sigma}$  with  $\sigma = \rho\beta$ . The constant  $s^{-\rho}$  can be absorbed into  $D(\cdot)$ , yielding the isoelastic form.

We note that the assumption of multiplicative costs cannot be dropped from [Theorem 2](#). [Example 2](#) in [Appendix C](#) shows that a non-multiplicative cost can have constant waste even though  $-\partial \ln C(a, \theta)/\partial \ln V(\theta)$  varies with  $\theta$ .

## 4. A Signaling Tournament

We now apply the constant-waste formula to signaling in a tournament, such as workers competing for a job. Richer models of tournament-like signaling and matching have been studied by [Hoppe, Moldovanu and Sela \(2009\)](#) and [Hopkins \(2012\)](#); they do not focus on quantifying waste.

Specifically, consider a market modeled as a tournament among  $N \geq 2$  candidates for a single prize of value or size  $s > 0$  (e.g., a promotion or a job offer). Candidates' types are their private information, drawn independently from a common distribution  $F$  on  $[0, 1]$ . Candidates simultaneously choose their observable signaling actions. The prize is awarded to the candidate with the highest perceived type.

If candidate  $i$  is perceived as type  $\hat{\theta}_i$ , her probability of winning is  $\mathbb{P}(\hat{\theta}_i > \max_{j \neq i} \hat{\theta}_j)$ . In a symmetric separating equilibrium, each candidate's expected benefit is

$$V(\theta) = s (F(\theta))^{N-1}.$$

Under the uniform distribution,  $F(\theta) = \theta$ , the function simplifies to  $V(\theta) = s\theta^{N-1}$ ; this is an isoelastic benefit with  $\beta = N - 1$ .

Assuming isoelastic signaling costs with unit strain elasticity, i.e.,  $C(a, \theta) = D(a)/\theta$  for some  $D(\cdot)$ , [Theorem 1](#) immediately implies a constant waste ratio:

$$W_N(\theta) := \frac{N-1}{N}. \tag{6}$$

This expression is  $1/2$  when  $N = 2$ ; it increases in  $N$ ; and as  $N \rightarrow \infty$ , the waste ratio approaches 1. In other words, greater competition exacerbates waste, with the entire surplus dissipated in the limit.

**The Tullock Connection.** Strikingly, the formula (6) is exactly the rent dissipation rate in a symmetric Tullock lottery contest with  $N$  players (Tullock, 1980). There, each player  $i$  chooses effort  $x_i \geq 0$  to win a prize of size  $s > 0$ , with a linear cost and winning probability  $x_i / \sum_j x_j$ .<sup>9</sup> In the symmetric equilibrium, each player exerts effort  $x^* = s(N - 1)/N^2$ , and total effort is  $Nx^* = s(N - 1)/N$ , hence  $(N - 1)/N$  of the prize is dissipated.

This connection is noteworthy given the settings’ differences.<sup>10</sup> The Tullock lottery contest is a complete-information game with a given winning probability function, whereas our tournament features asymmetric information and signaling to an observer. Yet both reduce to the same fundamental structure: agents burn resources to obtain a fixed prize; the intensity of competition, governed by the number of players, ends up determining surplus dissipation. Equilibrium is determined by a first-order condition that balances marginal cost against marginal benefit, with the formula (6) emerging in both.

**Generalizations and Differences.** For an arbitrary strain elasticity  $\sigma > 0$  and power distribution  $F(\theta) = \theta^k$  with  $k > 0$  (which yields benefit elasticity  $\beta = k(N - 1)$ ), Theorem 1 implies that the waste ratio (6) generalizes to  $k(N - 1)/(k(N - 1) + \sigma)$ . Tullock (1980) showed that in a contest with winning probability  $(x_i)^r / (\sum_j x_j)^r$  for  $r \in (0, 1]$ , the rent dissipation rate is  $r(N - 1)/N$ . We see that when  $r \neq 1$ , there is an important difference in large markets. As  $N \rightarrow \infty$ , the rent dissipation rate in the Tullock contest asymptotes to  $r$ , whereas in the signaling tournament it goes to 1 regardless of  $\sigma$  and  $k$ . Intuitively, the noise in a non-lottery Tullock contest can preserve some surplus even with extreme competition; but separating from a dense field of competitors forces full dissipation under signaling.

The two models also differ in their sensitivity to the cost structure. Consider a Tullock lottery contest (so  $r = 1$ ) with an isoelastic cost of effort,  $D(x) = x^\gamma$  with  $\gamma \geq 1$ . As this contest is isomorphic to one with a linear cost but winning probability parameter  $r = 1/\gamma$ , the rent dissipation rate is now  $(1/\gamma)(N - 1)/N$ . Greater cost convexity (higher  $\gamma$ ) reduces rent dissipation by dampening the marginal return to effort. By contrast, Theorem 1 implies that in the signaling tournament, the difficulty  $D(\cdot)$  is irrelevant; separation forces agents to scale their efforts with  $D(\cdot)$  exactly enough to leave waste unchanged. Unlike in a contest, then, signaling tournaments’ waste cannot be reduced by simply making better performance costlier (increasing  $\gamma$ ); one must reduce the number of competitors or flatten the prize

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<sup>9</sup> The “lottery” descriptor for the Tullock contest refers to effort entering the winning probability linearly in both the numerator and denominator. Nitzan (1994) and Corchón (2007) survey generalizations of this and many other aspects of contests.

<sup>10</sup> While Tullock contests have been mapped to various environments (e.g., Baye and Hoppe, 2003), we are not aware of any work relating them to the surplus dissipation in signaling.

gradient (the benefit elasticity  $\beta$ ).

## 5. Conclusion

We have proposed the **waste ratio**—the fraction of a type’s surplus dissipated through signaling (in the separating equilibrium)—as a natural measure of signaling inefficiency. Our main result, [Theorem 1](#), is a simple formula for this waste ratio in isoelastic environments:  $\beta/(\beta + \sigma)$ , where  $\beta$  is the elasticity of benefits and  $\sigma$  measures signaling strain, or higher types’ comparative advantage. This ratio is invariant to the type distribution, the stakes in signaling (a scale parameter of benefits), or the signaling difficulty (the type-independent scale or shape of the cost of higher signals).

The invariance to difficulty undermines common intuitions about the inefficiency of signaling. Consider the example of college admissions. There are debates about the difficulty of entrance exams, with recent trends favoring shorter, less complex tests (such as the digital SAT) to reduce student stress. Conversely, critics of grade inflation often call for harder exams to restore selectivity. Our results suggest that, when viewed through the canonical signaling lens, neither approach may address the underlying waste. Adjusting the difficulty of the test—whether making it easier or harder—need not change the total resource dissipation; it could merely rescale equilibrium effort while leaving the waste ratio constant.<sup>11</sup> To meaningfully reduce waste, policies should instead target the market structure itself. As long as admissions at selective colleges resemble winner-take-all signaling tournaments with many competitors, the process is likely to dissipate significant surplus, regardless of how the testing technology is calibrated.

While difficulty is irrelevant, the strain elasticity  $\sigma$  matters. Another approach, then, to reducing waste would be to make tests more discriminating in the sense of increasing  $\sigma$ . Waste would vanish as the elasticity  $\sigma \rightarrow \infty$ . Interestingly, this echoes a discussion in the biological signaling literature: animals can reliably convey information while incurring minimal waste ([Enquist, 1985](#)). The mechanism stems from sharply different marginal costs across types—originally proposed by [Zahavi \(1977\)](#) to refine his earlier “handicap” hypothesis—rather than difficulty. Scholars have argued that because Darwinian selection favors efficiency, it leads to biological signaling instruments that are cheap for high-quality types but prohibitive for low-quality types ([Penn and Számadó, 2020](#)); that corresponds to a high strain elasticity  $\sigma$ .

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<sup>11</sup> This invariance does rely on a single-dimensional framework; it could break if students also differ in test-taking aptitude separate from underlying ability, which would lead to “muddled information” ([Frankel and Kartik, 2019](#)).

We close by commenting on three limitations of our analysis. First, we focus exclusively on separating equilibria. That is consistent with much of the literature’s emphasis, often justified by stability-based arguments (e.g., [Cho and Sobel, 1990](#)). But there are also critiques of the exclusive focus on separating equilibria (e.g., [Mailath, Okuno-Fujiwara and Postlewaite, 1993](#)). Equilibria with some pooling, where certain types choose identical actions, can reduce signaling costs and waste.

Second, we focus on the dissipative cost of information revelation. Signaling activities can, of course, sometimes be productive; see [fn. 1](#). We expect that even in a broader welfare calculus, our waste ratio is a useful input: the net social value must weigh signaling’s intrinsic benefit against its waste.

Third, [Theorem 2](#) indicates that the waste ratio is less tractable outside the isoelastic class. It will then generally vary with type—so aggregation will depend on the type distribution—and more nuanced properties of the parameters. We do not suggest that isoelasticity should be taken literally. Rather, we view it as focal by analogy to how CRRA utility is canonical not because preferences literally exhibit constant relative risk aversion, but because it affords tractable analyses and scale-free results. We hope the waste formula  $\beta/(\beta + \sigma)$  is a similarly useful benchmark for signaling’s welfare cost. A broader lesson is the invariance of waste to difficulty and stakes under multiplicative costs.

## A. Separating Equilibria

A (*mixed*) *strategy* for the agent is a measurable mapping  $\alpha : \Theta \rightarrow \Delta(\mathbb{R}_{\geq 0})$ , where  $\Delta(\mathbb{R}_{\geq 0})$  denotes the set of probability distributions over actions. A *pure strategy* is a strategy  $\alpha$  such that  $\alpha(\theta)$  has singleton support for all  $\theta$ ; we denote a pure strategy more simply by  $A : \Theta \rightarrow \mathbb{R}_{\geq 0}$ . Since a belief concentrated on type 0 is the “worst belief” (by monotonicity of the benefit function  $V$ ), and hence is the most conducive off-path belief to support an equilibrium, we say that strategy  $\alpha$  defines a *separating equilibrium* if:

1. (Separation.) For any  $\theta \neq \theta'$ , the distributions are mutually singular:  $\alpha(\theta) \perp \alpha(\theta')$ .<sup>12</sup>
2. (Incentive compatibility.) For each  $\theta \in \Theta$ ,  $\alpha(\theta)$ -a.e.  $a$ , and all  $a' \in \mathbb{R}_{\geq 0}$ :

$$V(\theta) - C(a, \theta) \geq V(\hat{\theta}(a')) - C(a', \theta),$$

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<sup>12</sup> Recall that two distributions are mutually singular if each assigns probability one to a set that the other assigns probability zero. In other words, the two types choose distinct actions with probability one.

where  $\hat{\theta}(a') = \theta'$  if  $a' \in \text{supp}(\alpha(\theta'))$  for a unique  $\theta'$ , and  $\hat{\theta}(a') = 0$  otherwise.<sup>13</sup>

**Proposition 1.** *Any separating equilibrium is a pure-strategy equilibrium. Moreover, its strategy  $A : \Theta \rightarrow \mathbb{R}_{\geq 0}$  is continuous and strictly increasing on  $[0, \bar{\theta}]$ , continuously differentiable on  $(0, \bar{\theta})$ , and satisfies*

$$A'(\theta) = \frac{V'(\theta)}{C_a(A(\theta), \theta)} \quad \text{for } \theta > 0, \quad (7)$$

with boundary condition  $A(0) = 0$ .

Although we are not aware of an existing result that directly implies [Proposition 1](#), the proof follows familiar lines (cf. [Mailath, 1987](#)) and is provided in the [Supplementary Appendix](#). It is worth noting that because [Proposition 1](#) only establishes necessary conditions for a separating equilibrium, it does not require all of [Assumption 1](#); in particular, it is enough that  $V$  is  $\mathcal{C}^1$  rather than  $\mathcal{C}^2$ , and there is no need for differentiability of  $C_a$  or that  $C_{a\theta} < 0$ . Those three properties are instead used to verify sufficiency in the next result, whose proof—which largely follows [Mailath \(1987\)](#)—is also in the [Supplementary Appendix](#).<sup>14</sup>

**Proposition 2.** *A continuous function  $A : \Theta \rightarrow \mathbb{R}_{\geq 0}$  that is differentiable on  $(0, \bar{\theta})$  and satisfies (7) and  $A(0) = 0$  constitutes a separating equilibrium.*

## B. Proof of Theorem 2

**Necessity.** In a separating equilibrium with strategy  $A$ , type  $\theta$ 's payoff is  $U(\theta) := V(\theta) - C(A(\theta), \theta)$ . Observe that

$$U'(\theta) = -C_\theta(A(\theta), \theta) = -D(A(\theta))S'(\theta), \quad (8)$$

where the first equality is by the Envelope Theorem, and the second equality is by (2). Note that  $U'(\theta) > 0$  for  $\theta > 0$  because on this domain  $S'(\theta) < 0$  and  $A(\theta) > 0$  and hence  $D(A(\theta)) > 0$ . Consequently, for any  $\theta > 0$ , it holds that  $U(\theta) > 0$  (as  $U(0) = 0$ ) and also  $U(\theta) < V(\theta)$  (as  $C(A(\theta), \theta) > 0$  because  $A(\theta) > 0$ ).

<sup>13</sup> Note that the belief is stipulated as zero for any action in the support of multiple types' mixtures. This is for convenience; it is justified because Bayes' rule (formally,  $\hat{\theta}(\cdot)$  being the expectation under a regular conditional distribution) only pins down beliefs  $\mu$ -a.e., where  $\mu$  is the marginal distribution over actions induced by the prior  $F$  and strategy  $\alpha$ . Mutual singularity from point 1 of the definition ensures that for each  $\theta$  and  $\alpha(\theta)$ -a.e. action  $a$ , the belief  $\hat{\theta}(a) = \theta$ .

<sup>14</sup> In fact, even [Proposition 2](#) does not require  $V$  to be  $\mathcal{C}^2$ ; it only uses twice differentiability. [Assumption 1](#) part 1 states “twice continuously differentiable” for simplicity in lieu of “continuously differentiable, and twice differentiable”.

Suppose  $W(\theta) = k$  is constant. Then the [definition](#) of  $W$  and [Equation 2](#) combine to imply

$$D(A(\theta))S(\theta) = kV(\theta) \quad (9)$$

and

$$U(\theta) = (1 - k)V(\theta). \quad (10)$$

Note that  $k \in (0, 1)$  because  $U(\theta) \in (0, V(\theta))$  for  $\theta > 0$ .

Differentiating [\(10\)](#) yields

$$U'(\theta) = (1 - k)V'(\theta). \quad (11)$$

Equating the expressions in [\(8\)](#) and [\(11\)](#) gives

$$-D(A(\theta))S'(\theta) = (1 - k)V'(\theta),$$

into which we substitute for  $D(A(\theta))$  from [\(9\)](#) to obtain

$$-\frac{kV(\theta)}{S(\theta)}S'(\theta) = (1 - k)V'(\theta),$$

or after rearranging,

$$-\frac{S'(\theta)}{S(\theta)} \cdot \frac{V(\theta)}{V'(\theta)} = \frac{1 - k}{k}.$$

The left-hand side is  $-d \ln S / d \ln V$ . The result follows by letting  $\rho = (1 - k)/k > 0$ .

**Sufficiency.** Suppose [\(5\)](#) holds for some  $\rho > 0$ . Integrating yields  $S(\theta) = \kappa \cdot V(\theta)^{-\rho}$  for some  $\kappa > 0$ . Following the proof of [Theorem 1](#), the waste ratio is

$$W(\theta) = \frac{S(\theta)}{V(\theta)} \int_0^\theta \frac{V'(t)}{S(t)} dt = \frac{\kappa (V(\theta))^{-\rho}}{V(\theta)} \int_0^\theta \frac{V'(t)}{\kappa V(t)^{-\rho}} dt = (V(\theta))^{-(1+\rho)} \int_0^\theta V'(t) V(t)^\rho dt.$$

Using  $V(0) = 0$ , we evaluate the integral

$$\int_0^\theta V'(t) V(t)^\rho dt = \frac{(V(\theta))^{1+\rho}}{1 + \rho}.$$

Hence  $W(\theta) = 1/(1 + \rho)$ . □

## C. A Non-Multiplicative Cost Example

**Example 2.** Let  $V(\theta) = s\theta$  for  $s > 0$ , and  $C(a, \theta) = a^2/\theta + a^3/\theta^2$ . These functions satisfy [Assumption 1](#) but the cost is not multiplicatively separable. Moreover, by contrast to [Equation 5](#), we have

$$-\frac{\partial \ln C(a, \theta)}{\partial \ln V(\theta)} = \frac{\theta + 2a}{\theta + a},$$

which varies with  $\theta$ . We show that the waste ratio depends on stakes  $s$ , but is nevertheless constant across types.

Let  $c > 0$  be the unique solution to  $2c^2 + 3c^3 = s$ , and consider the strategy  $A(\theta) = c\theta$ . This linear (hence continuous and differentiable) separating strategy satisfies  $A(0) = 0$  and the differential equation [\(7\)](#), because

$$C_a(A(\theta), \theta) \cdot A'(\theta) = \left( \frac{2c\theta}{\theta} + \frac{3c^2\theta^2}{\theta^2} \right) \cdot c = 2c^2 + 3c^3 = s = V'(\theta).$$

Hence, by [Proposition 2](#),  $A$  constitutes a separating equilibrium.

The waste ratio is

$$W \equiv W(\theta) = \frac{C(A(\theta), \theta)}{V(\theta)} = \frac{c^2\theta + c^3\theta}{(2c^2 + 3c^3)\theta} = \frac{1 + c}{2 + 3c},$$

which is constant across types but depends on  $s$  because  $c$  varies with  $s$ . In particular, as  $s \rightarrow 0$ , we have  $c \rightarrow 0$  and  $W \rightarrow 1/2$ ; as  $s \rightarrow \infty$ , we have  $c \rightarrow \infty$  and  $W \rightarrow 1/3$ .  $\diamond$

## References

- Baye, Michael R and Heidrun C Hoppe, “The Strategic Equivalence of Rent-seeking, Innovation, and Patent-race Games,” *Games and Economic Behavior*, 2003, 44 (2), 217–226.
- Bénabou, Roland and Jean Tirole, “Incentives and Prosocial Behavior,” *American Economic Review*, December 2006, 96 (5), 1652–1678.
- Bernheim, B. Douglas and Aaron L. Bodoh-Creed, “Pervasive Signaling,” *Theoretical Economics*, 2023, 18 (1), 163–196.
- Cho, In-Koo and Joel Sobel, “Strategic Stability and Uniqueness in Signaling Games,” *Journal of Economic Theory*, 1990, 50 (2), 381–418.
- Corchón, Luis C, “The Theory of Contests: a Survey,” *Review of Economic Design*, 2007, 11 (2), 69–100.
- Enquist, Magnus, “Communication During Aggressive Interactions with Particular Reference to Variation in Choice of Behaviour,” *Animal Behaviour*, 1985, 33 (4), 1152–1161.
- Frankel, Alex and Navin Kartik, “Muddled Information,” *Journal of Political Economy*, August 2019, 129 (4), 1739–1776.
- Fudenberg, Drew and Jean Tirole, *Game Theory*, Cambridge, MA: MIT Press, 1991.
- Hopkins, Ed, “Job Market Signaling of Relative Position, or Becker Married to Spence,” *Journal of the European Economic Association*, 2012, 10 (2), 290–322.
- Hoppe, Heidrun C., Benny Moldovanu, and Aner Sela, “The Theory of Assortative Matching Based on Costly Signals,” *Review of Economic Studies*, 2009, 76 (1), 253–281.
- Kartik, Navin, “Strategic Communication with Lying Costs,” *Review of Economic Studies*, October 2009, 76 (4), 1359–1395.
- Koutsoupias, Elias and Christos Papadimitriou, “Worst-case equilibria,” in “Proceedings of the 16th Annual Symposium on Theoretical Aspects of Computer Science (STACS)” Springer 1999, pp. 404–413.
- Mailath, George, “Incentive Compatability in Signaling Games with a Continuum of Types,” *Econometrica*, 1987, 55 (6), 1349–1365.



- , Masahiro Okuno-Fujiwara, and Andrew Postlewaite, “Belief-Based Refinements in Signalling Games,” *Journal of Economic Theory*, August 1993, 60 (2), 241–276.
- Nitzan, Shmuel, “Modelling Rent-seeking Contests,” *European Journal of Political Economy*, 1994, 10, 41–60.
- Nöldeke, Georg and Larry Samuelson, “How Costly is the Honest Signaling of Need?,” *Journal of Theoretical Biology*, 1999, 197 (4), 527–539.
- Penn, Dustin J. and Szabolcs Számádó, “The Handicap Principle: How an Erroneous Hypothesis became a Scientific Principle,” *Biological Reviews*, 2020, 95 (1), 267–290.
- Riley, John G., “Silver Signals: Twenty-Five Years of Screening and Signaling,” *Journal of Economic Literature*, 2001, 39 (2), 432–478.
- Roughgarden, Tim, *Selfish Routing and the Price of Anarchy*, MIT Press, 2005.
- , Vasilis Syrgkanis, and Eva Tardos, “The Price of Anarchy in Auctions,” *Journal of Artificial Intelligence Research*, 2017, 59, 59–101.
- Sobel, Joel, “Signaling Games,” in Robert A. Meyers, ed., *Encyclopedia of Complexity and Systems Science*, Springer, 2009, pp. 8116–8143.
- Spence, Michael, “Job Market Signaling,” *Quarterly Journal of Economics*, August 1973, 87 (3), 355–374.
- Tullock, Gordon, “Efficient Rent Seeking,” in James M. Buchanan, Robert D. Tollison, and Gordon Tullock, eds., *Toward a Theory of the Rent-Seeking Society*, College Station: Texas A&M University Press, 1980, pp. 97–112.
- Zahavi, Amotz, “The Cost of Honesty (Further Remarks on the Handicap Principle),” *Journal of Theoretical Biology*, 1977, 67 (3), 603–605.

# Supplementary Appendix

## D. Omitted Proofs

The following lemma is used in the proof of [Proposition 1](#).

**Lemma 1.**  $C(\cdot, 0)$  is strictly increasing on  $\{a \geq 0 : C(a, 0) < \infty\}$ .

**Proof.** Since  $C(\cdot, \theta)$  is strictly increasing for  $\theta > 0$ , the limit definition of  $C(\cdot, 0)$  implies it is weakly increasing. To establish strict monotonicity, consider  $a'' > a' \geq 0$  with  $C(a'', 0) < \infty$ . For any  $\theta > 0$ , we have

$$C(a'', \theta) - C(a', \theta) = \int_{a'}^{a''} C_a(x, \theta) dx.$$

For any  $x > 0$ , the function  $C_a(x, \cdot)$  is strictly decreasing on  $(0, \bar{\theta}]$  because  $C_{a\theta} < 0$  on this domain, and so  $L(x) := \lim_{\theta \downarrow 0} C_a(x, \theta)$  exists in  $\mathbb{R}_{>0} \cup \{\infty\}$ . By monotone convergence,

$$C(a'', 0) - C(a', 0) = \int_{a'}^{a''} L(x) dx > 0. \quad \square$$

**Proof of [Proposition 1](#).** We proceed in three steps.

STEP 1: ANY SEPARATING EQUILIBRIUM HAS A PURE STRATEGY.

Consider any separating equilibrium strategy  $\alpha$  and type  $\theta$ . Incentive compatibility requires  $C(a, \theta) = C(a', \theta)$  for  $\alpha(\theta)$ -a.e.  $a$  and  $a'$ , since any such actions induce the same belief  $\theta$ . By the strict monotonicity of  $C(\cdot, \theta)$  for  $\theta > 0$  and using [Lemma 1](#) for  $\theta = 0$  (noting that this type will never choose an action with infinite cost), we have  $a = a'$  for  $\alpha(\theta)$ -almost every  $a$  and  $a'$ , which implies  $\alpha(\theta)$  has singleton support. Hence  $\alpha$  is a pure strategy.

STEP 2: THE BOUNDARY CONDITION  $A(0) = 0$ .

Let  $A$  be a separating equilibrium (pure) strategy. If  $A(0) > 0$ , then  $V(0) - C(A(0), 0) < V(0) - C(0, 0)$ , contradicting incentive compatibility (IC) for type 0.

STEP 3: CONTINUITY, DIFFERENTIABILITY, AND MONOTONICITY.

Let  $A$  be a separating equilibrium (pure) strategy. We first show  $A$  is continuous. Consider any  $\theta^* \in (0, \bar{\theta})$  and let  $a^- := \lim_{\theta \uparrow \theta^*} A(\theta)$  and  $a^+ := \lim_{\theta \downarrow \theta^*} A(\theta)$  (passing to subsequences if necessary, noting that  $A$  is bounded on compact subsets by IC). IC for type  $\theta^*$

implies that for all  $\theta < \theta^*$ , we have

$$V(\theta^*) - C(A(\theta^*), \theta^*) \geq V(\theta) - C(A(\theta), \theta^*).$$

Taking  $\theta \uparrow \theta^*$  and using continuity of  $V$  and  $C$  yields

$$V(\theta^*) - C(A(\theta^*), \theta^*) \geq V(\theta^*) - C(a^-, \theta^*). \quad (12)$$

Applying IC in the reverse direction (for a type  $\theta < \theta^*$  to not mimic  $\theta^*$ ) and taking the same limit yields the opposite inequality to (12). Hence  $C(A(\theta^*), \theta^*) = C(a^-, \theta^*)$ . An analogous argument using types above  $\theta^*$  gives  $C(A(\theta^*), \theta^*) = C(a^+, \theta^*)$ . Since  $C(\cdot, \theta^*)$  is strictly increasing, this implies  $a^- = A(\theta^*) = a^+$ , so  $A$  is continuous at  $\theta^*$ . The same argument, but using only  $a^-$  or  $a^+$  as applicable, also establishes continuity when  $\theta^* \in \{0, \bar{\theta}\}$ .

We now establish differentiability. Fix  $\theta > 0$  and  $\theta' \neq \theta$ . Adding the IC inequalities for  $\theta$  to not mimic  $\theta'$  and vice-versa, and rearranging, yields

$$C(A(\theta'), \theta) - C(A(\theta), \theta) \geq V(\theta') - V(\theta) \geq C(A(\theta'), \theta') - C(A(\theta), \theta'). \quad (13)$$

Since  $C$  is differentiable in its first argument, the mean value theorem implies that the left-hand side of (13) equals  $C_a(\tilde{a}, \theta) \cdot (A(\theta') - A(\theta))$  for some  $\tilde{a}$  between  $A(\theta)$  and  $A(\theta')$ , and analogously for the right-hand side with  $C_a(\tilde{a}', \theta')$  for some  $\tilde{a}'$  between  $A(\theta)$  and  $A(\theta')$ . Substituting into (13) and dividing by  $\theta' - \theta$  gives

$$C_a(\tilde{a}, \theta) \left( \frac{A(\theta') - A(\theta)}{\theta' - \theta} \right) \geq \frac{V(\theta') - V(\theta)}{\theta' - \theta} \geq C_a(\tilde{a}', \theta') \left( \frac{A(\theta') - A(\theta)}{\theta' - \theta} \right), \quad (14)$$

where the inequalities are written for  $\theta' > \theta$  and would flip if  $\theta' < \theta$ . Either way, take  $\theta' \rightarrow \theta$ : since  $\tilde{a}, \tilde{a}' \rightarrow A(\theta)$  by continuity of  $A$ , and  $C_a$  is continuous, both  $C_a(\tilde{a}, \theta)$  and  $C_a(\tilde{a}', \theta')$  converge to  $C_a(A(\theta), \theta) > 0$ . Since the middle term of (14) converges to  $V'(\theta)$  and the two outer terms share the common factor  $\frac{A(\theta') - A(\theta)}{\theta' - \theta}$  with coefficients converging to the same positive limit, we conclude that  $A'(\theta)$  exists with

$$A'(\theta) = \frac{V'(\theta)}{C_a(A(\theta), \theta)},$$

which is continuous on  $(0, \bar{\theta})$  because  $V'$ ,  $C_a$ , and  $A$  are all continuous on that domain and  $C_a(A(\theta), \theta) > 0$  for  $\theta > 0$  (noting that  $A(\theta) > 0$  for  $\theta > 0$  by separation).

Finally, since we have established  $A'(\theta) > 0$  for  $\theta > 0$  and that  $A$  is continuous, it follows

that  $A$  is strictly increasing on  $[0, \bar{\theta}]$ .  $\square$

**Proof of Proposition 2.** Let  $A$  have the stated properties. Note that for  $\theta > 0$ , if  $A(\theta) = 0$  then  $C_a(0, \theta) > 0$ : the derivative must exist and be nonzero to satisfy (7), as  $V'(\theta) > 0$ , and the derivative cannot be negative because  $C(\cdot, \theta)$  is continuous and  $C_a(a, \theta) > 0$  for  $a > 0$ . Using Assumption 1, it follows that for  $\theta > 0$ , no matter the value of  $A(\theta)$ , both  $V'(\theta) > 0$  and  $C_a(A(\theta), \theta) > 0$ , and hence (7) implies  $A'(\theta) > 0$ . By continuity of  $A$ , it is strictly increasing on  $[0, \bar{\theta}]$ . Moreover,  $A$  is twice differentiable on  $(0, \bar{\theta})$  since the right-hand side of (7) is differentiable in  $\theta$  (because Assumption 1 entails  $V$  twice differentiable and  $C_a$  differentiable on the relevant domain).

We now verify that  $A$  defines a separating equilibrium. Since any off-path action is met with belief  $\hat{\theta} = 0$ , it is strictly worse than action 0. So any type  $\theta$  can be viewed as only choosing which type  $\tilde{\theta}$  to mimic, with payoff

$$\Pi(\theta, \tilde{\theta}) := V(\tilde{\theta}) - C(A(\tilde{\theta}), \theta).$$

For any  $\theta > 0$ , the first order-condition obviously holds, since  $A$  solves (7). Let us verify the second-order condition. Using numeric subscripts for partial derivatives of  $\Pi$  in the usual way, and restricting attention to the domain  $\theta > 0$ , we have

$$\Pi_{22}(\theta, \tilde{\theta}) = V''(\tilde{\theta}) - C_{aa}(A(\tilde{\theta}), \theta) \cdot (A'(\tilde{\theta}))^2 - C_a(A(\tilde{\theta}), \theta) \cdot A''(\tilde{\theta}). \quad (15)$$

Totally differentiating (7) with respect to  $\theta$  yields

$$V''(\theta) = C_{aa}(A(\theta), \theta) \cdot (A'(\theta))^2 + C_{a\theta}(A(\theta), \theta) \cdot A'(\theta) + C_a(A(\theta), \theta) \cdot A''(\theta). \quad (16)$$

Substituting from (16) into (15) and then evaluating at  $\tilde{\theta} = \theta$  yields

$$\Pi_{22}(\theta, \theta) = C_{a\theta}(A(\theta), \theta) \cdot A'(\theta) < 0,$$

where the inequality follows from  $A'(\theta) > 0$  and, by Assumption 1,  $C_{a\theta} < 0$  on  $\mathbb{R}_{>0} \times (0, \bar{\theta})$ . This verifies the second-order condition and hence local optimality for each type  $\theta > 0$ . It follows from the single-crossing assumption  $C_{a\theta} < 0$  that (global) incentive compatibility holds on the domain  $(0, \bar{\theta})$ , i.e., for any pair of types in this domain, neither wants to mimic the other.

Finally, we address type 0. Consider an arbitrary other type  $\theta > 0$ . We must show  $\Pi(\theta, \theta) \geq \Pi(\theta, 0)$  and  $\Pi(0, 0) \geq \Pi(0, \theta)$ . Taking each in turn:

1. Incentive compatibility on  $(0, \bar{\theta})$  implies  $\Pi(\theta, \theta) \geq \Pi(\theta, \tilde{\theta})$  for all  $\tilde{\theta} > 0$ , while continuity of  $V$ ,  $A$  and  $C(\cdot, \theta)$  imply  $\Pi(\theta, \tilde{\theta}) \rightarrow \Pi(\theta, 0)$  as  $\tilde{\theta} \rightarrow 0$ . Hence,  $\Pi(\theta, \theta) \geq \Pi(\theta, 0)$ .
2. For any  $\tilde{\theta} > 0$ , the previous point implies  $\Pi(\tilde{\theta}, \tilde{\theta}) \geq \Pi(\tilde{\theta}, 0) = 0$ , and hence  $0 \leq C(A(\tilde{\theta}), \tilde{\theta}) \leq V(\tilde{\theta})$ . Since  $V(\tilde{\theta}) \rightarrow 0$  as  $\tilde{\theta} \rightarrow 0$ , it follows that  $\Pi(\tilde{\theta}, \tilde{\theta}) \rightarrow 0 = \Pi(0, 0)$ . Moreover, for any  $\tilde{\theta} > 0$  we have  $\Pi(\tilde{\theta}, \tilde{\theta}) \geq \Pi(\tilde{\theta}, \theta)$  by incentive compatibility on  $(0, \bar{\theta})$ , and  $\Pi(\tilde{\theta}, \theta) \rightarrow \Pi(0, \theta)$  by the limit property of  $C(\cdot, 0)$  in [Assumption 1](#). Hence  $\Pi(0, 0) \geq \Pi(0, \theta)$ . □